

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 10



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Recap of Previous Lecture



Topic

Bell number



Topic

Number of equivalence relations



Topic

Partial Order Relation and Partially Ordered Set



Topic

Totally Ordered Set and Total Order Relation



Topics to be Covered



Topic

Least Upper Bound



Topic

Greatest Lower Bound



Topic

E.g. of Least Upper Bound & Greatest Lower Bound



eg. $\rightarrow$ Let $A_1 = \{1, 2, 3, 4\}$
and $(A_1, \div)$ is a POSET

$\rightarrow$ We know, that there are some pair of elements in set A_1 that are not comparable w.r.t. POSET $(A_1, \div)$

$\therefore (A_1, \div)$ is just a POSET, but not a TOSET.

$\rightarrow$ eg. Let $A_2 = \{1, 2, 4, 8\}$
and $(A_2, \div)$ is a POSET.

$\rightarrow$ Every pair of elements of set A_2 are comparable $\therefore (A_2, \div)$ is a TOSET

$\rightarrow (A_2, \div)$ is a TOSET, but it is also a POSET.

Note:-

① If A is any set of real numbers then $(A, \leq)$ is always a "TOSET"

② If A is any set of positive integers then $(A, \div)$ may or may not be a TOSET.

eg. $C_1 = \{\emptyset, \{1\}, \{1, 2\}\}$ ③ If C is any Collection of sets, then $(C, \subseteq)$ may or may not be a TOSET

$C_2 = \{\emptyset, \{1\}, \{2\}\}$

Here, $(C_1, \subseteq)$ is a TOSET.
but $(C_2, \subseteq)$ is not a TOSET.

Note:-→

If R is a partial order relation, and

$$a \overset{R}{\prec} b, \text{ i.e. } (a, b) \in R$$

then
we say that

'a' is
at lower
side



'b' is at
upper side



$(a, b) \in R$

element
at lower
side

element at
upper side

eg: $A = \{1, 2, 3\}$
and $(A, \leq)$ is a poset,

We know, $(1, 2) \in \leq$
 $\therefore$ '1' at lower side
& '2' at upper side

eg: Let $A = \{1, 2, 3, 4\}$
and $(A, \geq)$ is a poset.

* given 2 & 4 are two
elements in the above poset
then $4 \geq 2$

$\therefore (4, 2) \in \geq$

Hence, '4' is at lower side
& '2' is at upper side



Topic : Least Upper Bound

/ lub / Join / Supremum



Let $(A, \leq)$ be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$a \leq c$ and $b \leq c$,

→ then c is called upper bound of a and b ,

ie, $(a, c) \in \leq$ and $(b, c) \in \leq$

ie, ' c ' is at the upper side of both the elements a & b

⊗ And if there exists no ^{other} element $d \in A$ such that

$a \leq d$ and $b \leq d$ and $d \leq c$,

Note: ($d \neq c$)

$(a, d) \in \leq$ and $(b, d) \in \leq$ (and) $(d, c) \in \leq$

then c is called least upper bound of a and b

d is the upper bound of a & b

f

d is at lower side of c

then ' c ' is least of all upper bounds of a & b



Topic : Least Upper Bound

/ lub / Join / Supremum



Let (A, R) be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$a R c$ and $b R c$,

→ then c is called upper bound of a and b ,

ie, $(a, c) \in R$ and $(b, c) \in R$

ie, 'c' is at the upper side of both the elements a & b

⊗ And if there exists no element $d \in A$ such that

$a R d$ and $b R d$ and $d R c$,

other Note: $(d \neq c)$

$(a, d) \in R$ and $(b, d) \in R$ (and) $(d, c) \in R$

then c is called least upper bound of a and b

d is the upper bound of a & b

f

d is at lower side of c

then 'c' is least of all upper bounds of a & b

* Least upper bounds of two elements a & b
Can be denoted by $\text{lub}(a,b)$ or $a \vee b$

$$\text{lub}(a,b) = \text{lub}(b,a)$$

Notation for
least upper bound



Topic : Greatest Lower Bound

/ glb / Meet / Infimum



Let $(A, \leq)$ be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$$c \leq a \text{ and } c \leq b,$$

then c is called lower bound of a and b ,

$$(c, a) \in \leq \text{ and } (c, b) \in \leq$$

' c ' is at lower side of both the elements a & b

And if there exists no element $d \in A$ such that

$$d \leq a \text{ and } d \leq b \text{ and } c \leq d, \quad \text{other } (d \neq c)$$

$(d, a) \in \leq \neq (d, b) \in \leq$ and $(c, d) \in \leq$ then c is called greatest lower bound of a and b

' d ' is the lower bound of a and b & ' d ' is at upper side of c , then ' c ' is the greatest lower bound of a & b



Topic : Greatest Lower Bound

/ glb / Meet / Infimum



Let (A, R) be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$c R a$ and $c R b$,

then c is called lower bound of a and b ,

$(c, a) \in R$ and $(c, b) \in R$

' c ' is at lower side of both the elements a & b

And if there exists no element $d \in A$ such that

$d R a$ and $d R b$ and $c R d$,

other

$(d \neq c)$

$(d, a) \in R$ & $(d, b) \in R$ and $(c, d) \in R$ then c is called greatest lower bound of a and b

' d ' is the lower bound of a and b & ' d ' is at upper side of c , then ' c ' is the greatest lower bound of a & b

Note:- Greatest lower bound of two elements a & b
Can be denoted by $glb(a, b)$ or $a \wedge b$

↖ Notation
for glb

$$glb(a, b) = glb(b, a)$$

eg: Let $A = \{1, 2, 3, 4\}$, and $(A, \leq)$ is a POSET

then: $\text{lub}(2, 3) = ?$

$\text{glb}(2, 3) = ?$

$$2 \neq 1 \quad 2 \leq 2$$

$$3 \neq 1 \quad 3 \neq 2$$

$$2 \leq 3$$
$$3 \leq 3$$

$$2 \leq 4$$
$$3 \leq 4$$

3 & 4 are at the upper side of both 2 & 3

∴ 3 & 4 both are upper bounds of 2 & 3

But $3 \leq 4$ & $4 \not\leq 3$

∴ least among 3 & 4 is 3

Hence $\text{lub}(2, 3) = 3$

eg: Let $A = \{1, 2, 3, 4\}$, and $(A, \leq)$ is a POSET

then: $\text{lub}(2, 3) = ?$

$\text{glb}(2, 3) = ?$

$1 \leq 2$	$2 \leq 2$	$3 \not\leq 2$	$4 \not\leq 2$
$1 \leq 3$	$2 \leq 3$	$3 \leq 3$	$4 \not\leq 3$

Elements 1 & 2 are } $\Rightarrow \therefore 1 \& 2$ } $\Rightarrow$ But $1 \leq 2 \& 2 \not\leq 1$
At the lower side } are the } $\therefore 2$ is greatest
of both 2 & 3 } lower bounds } Among 1 & 2
of 2 & 3 } of 2 & 3 } Hence $\text{glb}(2, 3) = 2$

eg: Let $A = \{1, 2, 3, 4\}$, and $(A, \leq)$ is a POSET

then: $\text{lub}(2, 3) = 3$

$\text{glb}(2, 3) = 2$

Note:- If A is any set of real numbers and $(A, \leq)$ is a POSET, then for two elements $a, b \in A$

$$\text{lub}(a, b) = \text{Max}(a, b)$$

$$\text{glb}(a, b) = \text{Min}(a, b)$$

Note :- $\rightarrow$

Let A is a set of all positive integers
and $(A, \div)$ is a POSET.

for two elements $a, b \in A$

$$\text{lub}(a, b) = \text{LCM}(a, b)$$

$$\text{glb}(a, b) = \text{GCD}(a, b)$$

Note: →

$P(A)$ is
a Collection
of sets



↖ A set which
is superset of
both X & Y is an upper bound
of X & Y

Let A is any finite set and $P(A)$ is
the power set of set A , then $(P(A), \subseteq)$
is a POSET.

For two elements (sets) $X, Y \in P(A)$

$$\text{lub}(X, Y) = X \cup Y$$

$$\text{glb}(X, Y) = X \cap Y$$

Q:- Let $A = \{2, 3, 4\}$ and $(A, \div)$ is a POSET

then $\text{lub}(2, 4) = 4$

$$\text{glb}(2, 4) = 2$$

Q:- Let $A = \{2, 3, 4\}$ and $(A, \div)$ is a POSET

then $\text{lub}(2, 3) = \cancel{6} = \text{Does not exist}$

$\text{glb}(2, 3) = \cancel{1} = \text{Does not exist}$

Wrong

because 6 & 1
are not present
in the set A.

lub & glb for a pair of
elements if exists, then it will from the set itself.

Note :- $\rightarrow$

Let A is a set of all positive integers
and $(A, \div)$ is a POSET.

for two elements $a, b \in A$

$$\text{lub}(a, b) = \text{LCM}(a, b)$$

$$\text{glb}(a, b) = \text{GCD}(a, b)$$

It is the set of all positive integers
 $\therefore$ LCM and GCD
are guaranteed to
be present within
the set A .

Note:- If A is any set of Positive integers
and $(A, \div)$ is a POSET.

then for two elements $a, b \in A$

$\text{lub}(a, b) =$ Nothing can be said,
because we don't know about the elements
of the set.

$\text{glb}(a, b) =$ Nothing can be said,
because we don't know about the elements
of the set.

Note:-

Let C is any Collection of sets, then
 $(C, \subseteq)$ is a POSET

for two sets $X, Y \in C$

$$C = \{ \{1\}, \{2\} \}$$

$$\text{lub}(\{1\}, \{2\}) = \text{Does not exist}$$

$$\text{lub}(X, Y) = \text{Nothing can be said}$$

(We don't know the elements of the set C)

$$\text{glb}(X, Y) = \text{Nothing can be said}$$

(We don't know the elements of the set C)

Q: Let $A = \{1, 2, 3, 4, 24\}$ and $(A, \div)$
is a POSET,

$$\text{lub}(2, 3) = 24$$

$$\text{glb}(2, 3) = 1$$

Q: Let $A = \{2, 3, 4, 24\}$ and $(A, \div)$
is a POSET,

$$\text{lub}(2, 3) = 24$$

$$\text{glb}(2, 3) = \text{Does not exist}$$

Note :- →

In a POSET lub and/or glb
need not exist for Every pair
of elements



2 mins Summary



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Least Upper Bound

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Greatest Lower Bound

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E.g. of Least Upper Bound & Greatest Lower Bound

THANK - YOU